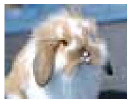
New shape discovered
A study has revealed the Scutoid, a new geometric shape used by nature to pack cells together. **https://dgit.in/sep18-75-1**



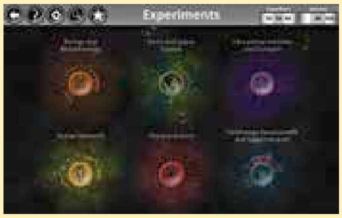
Cute bunny survey
Researchers are finding out exactly what features in rabbits make them cute, for safer breeding. **https://dgit.in/sep18-75-2**

for over a year in space. In previous astrobiology experiments, the microorganisms exposed to space always ended up dying, perhaps because the microorganisms were housed in thin films. The Tanpopo experiment had thicker layers of microorganisms. The outer layer of a species of bacteria, Deinococcus radiodurans died, but the dead organisms protected the inner layers, allowing them to survive for over a year.

CHONDRULE FORMATION

Fred Hoyle formulated the theory of stellar nucleosynthesis, which was a way to explain the origins of elements. This was the first scientific paper to show the elements that make up everything, including human bodies, were cooked in the hearts of stars that were long dead. There were those before him who had suspected the process, and from Carl Sagan to Moby, the idea is common in pop culture. While we can understand that the elements were produced within the stars, what remains unexplained is how this matter accumulated into clumps, which eventually become planets, moons and other stars. These clumps of matter, known as chondrules are the rudimentary particles that formed the building blocks of our solar system. The Experimental Chondrule Formation at the International Space Station (EXCISS) tests one of the many theories to explain chondrule formation – one of electrical discharges. The experiment simulates solar nebular conditions – the conditions before the formation of the solar system. Extremely fine silicate dust is transformed into chondrules through electric discharges, and the interaction of the dust particles are observed with a high definition camera. Some of the particles melt, while others do not, during the electrical discharges. One of the goals of the experiment is to understand how the molten particles behave differently from the unmolten particles. The experiment is controlled by a Raspberry-Pi.

SPACE STATION RESEARCH XPLORER



Thousands of experiments have been conducted on board the ISS, and you can explore all of them through an application dedicated to that purpose. The Space Station Research Xplorer allows you to browse through the experiments by categories, mission number, or sponsor. The application is available for free on Android and iOS.

SQUIRMING SPERM

NASA has sent up sea urchin, bull and human sperm to space to check how their behavior changes in a microgravity environment. The purpose of the experiment is to understand the feasibility of reproduction in deep space missions. Reproductive health deteriorates with age, and the data gathered during the experiments will help identify more approaches to tackle the decline in reproductive health with age, and improve the quality of life for humans on Earth. For the experiment, the frozen samples of the sperm were sent up to ISS, thawed out, and introduced to a chemical mixture that triggers activation of the sperm. The sperm was recorded on video, so that scientists have a record of the movement. Finally, the samples were mixed with preservatives and returned to the Earth. On Earth, the sperm samples were further analysed. Previous experiments have shown that the sperm get activated and start moving more quickly in microgravity, but the steps required for fusing with the egg is slowed down, or does not occur at all.

HUMANS AS A LABORATORY

A wide range of studies are conducted on the humans. On board the ISS, every single thing has to be planned to ensure the success of the mission. One of the experiments on board the space station, is to be called the Food Acceptability, Menu Fatigue, and Aversion in ISS Missions experiment. The purpose is to find out the effect of a repeated menu on board the ISS. If the astronauts are repeatedly given the same food, they may not consume the necessary intake to keep them healthy. Crew members on board the ISS often experience a loss in body mass, and menu fatigue could be one of the contributing factors.

Cerebral autoregulation is a process by which the brain maintains a healthy blood supply to itself, even when the heart and blood vessels cannot maintain a normal blood pressure. After their return to the Earth, many astronauts experience light headedness. This may be because of the changes in the blood flow to the brain during the spaceflight. The investigation monitors the blood pressure of the astronauts through the duration of the spaceflight, and afterwards. Early findings indicate that cerebral autoregulation is preserved in space, and perhaps even improved.

The Culture, Values, and Environmental Adaptation in Space experiment surveys those on the ISS on a regular basis, to understand changes in their values, culture, and strategies to adapt to the environment. Despite the exhaustive amount of training that the spacefarers go through, there are still conflicts and problems that have known to occur in space. The findings of the experiment suggests that a new “Space Culture” emerges, as a strategy for tackling the cultural differences and allows multinational teams to work together. Understanding the inner and interpersonal factors is important for future long duration space voyages, such as manned Mars missions. d